DI-725 : Transformers and Attention-Based Deep Networks Assignment - 2

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***Abstract*—We present a compact DETR-based object detector fine-tuned on the AU-AIR low-altitude UAV dataset. To accommodate limited computational resources, we froze the initial layers of the ResNet-50 backbone, except for the final (fourth) one. The original small DETR model outputs 92 classes; however, the AU-AIR dataset defines only 8. Therefore, we mapped the 8 AU-AIR classes into the 92-class output space. Input frames were resized from 1080×1920 to 800×800 pixels, as in the original DETR; ground-truth boxes in (ymin, xmin, height, width) format were converted to center format for regression and back to corner format for IoU computation. We trained the model using a combination of L1 and Generalized IoU loss for bounding boxes and CrossEntropy loss for classification.**

***Index Terms*—Transformers, DETR, Object Detection, AU-AIR, UAV Dataset, RES-NET-50 , Hybrid-Architectures**

I. Introduction

Object detection involves both localization and classification to identify and draw bounding boxes around each instance of an object class in an image. Aerial imagery—such as that captured by low-altitude UAVs—poses unique challenges: objects appear at varying scales, orientations, and densities, as noted in the AU-AIR research report. Traditional CNN-based detectors often struggle under such conditions.

The DEtection TRansformer (DETR) architecture has shown strong performance on natural image benchmarks by formulating detection as a set prediction task. However, DETR's high computational cost and memory demands limit its use on resource-constrained hardware. Additionally, AU-AIR is not a natural image dataset, making transfer learning more difficult. In this work, we explore three different architectures: CNN, Transformer, and a hybrid CNN-Transformer model, and conclude that a lightweight DETR-variant hybrid architecture performs best when fine-tuned on the AU-AIR traffic surveillance dataset.

II. Dataset

The AU-AIR dataset consists of 32,823 annotated video frames labeled with 8 traffic-related classes: *person, car, truck, van, motorbike, bicycle, bus, and trailer*. We also include a "no-object" background class. Annotations are provided in normalized *(ymin, xmin, height, width)* format. Following the setup in the AU-AIR report, we split the data into 30,000 frames for training/validation and 2,823 for testing.

During training, we convert bounding boxes from *(ymin, xmin, height, width)* to center format *(cx, cy, width, height)*.

III. Modeling

- Backbone: We use a ResNet-50 backbone and freeze all layers except the final one (layer4), reducing the number of trainable parameters from over 40 million to approximately 32 million. This adjustment was necessary to fit the model into a 4 GB GPU.
- Transformer: The architecture includes 6 encoder and 6 decoder layers, with 256 hidden dimensions, 8 attention heads, and 100 learned object queries.
- A linear classification head outputs 92 logits per query (91 classes + 1 "no-object" class).
- A linear bounding box head predicts normalized (cx, cy, w, h) coordinates, with a final sigmoid activation to constrain outputs to [0,1].
- Loss Function:
  - Hungarian Matcher: Used to optimally match predicted boxes with ground truth.
  - Bounding Boxes: L1 loss on (cx, cy, w, h) plus Generalized IoU loss on converted (xmin, ymin, xmax, ymax). However, the IoU loss proved unstable and was later removed.
  - Classification: CrossEntropyLoss applied to the raw logits.
  - Total loss = bbox_loss + cls_loss, equally weighted.

The code is available on my GitHub repository [2].

IV. Evaluation and Results

As reflected in our WANDB [1] tracking, the results were not sufficient to report meaningful mAP values. More time is required to identify an effective transfer learning strategy from natural image data to aerial imagery. Our goal was to report COCO-style mean Average Precision (mAP) scores over multiple IoU thresholds on the test set and to compare these

results against two lightweight models previously reported in the AU-AIR research report: YOLOv3-Tiny (mAP: 30.22%) and MobileNetV2-SSDLite (mAP: 19.50%).

Despite extensive experimentation—including training dozens of architectures and testing hundreds of parameter configurations—reliable results could not be obtained. No model was able to successfully overfit even a small subset of the dataset (e.g., 10 or 100 images). This limitation was primarily due to GPU memory constraints: the original image resolutions exceeded available resources, while resizing them significantly degraded the information necessary for accurate detection. Increasing model capacity to overfit the reduced images proved ineffective, as it neither resolved the overfitting issue nor alleviated memory limitations.

V. Discussion

- Convergence: Classification loss decreased quickly, while bounding box and total losses plateaued between 0.03 and 0.1 after 3–5 epochs. This may indicate limited model capacity or the difficulty of learning from downscaled inputs. Even small batches (2–3 images) triggered CUDA out-of-memory errors. Given that DETR was trained on natural images, fine-tuning on aerial imagery might not be an ideal transfer learning strategy. However, our DETR-based model outperformed all alternatives tested, including models trained from scratch.
- Trade-offs: We explored CNN-only, Transformer-only, and hybrid (CNN + Transformer) architectures. The hybrid model offered the best balance of performance and visual quality, likely because CNNs capture local features effectively while Transformers model global dependencies.
- Challenges:
  - Generalized IoU loss introduced instability, even when scaled by a small constant.
  - Image size significantly affected training—reducing resolution helped with memory but degraded box quality.
  - Due to time constraints, we could not explore more advanced strategies (e.g., two-phase learning). Our main focus remained on obtaining mAP metrics.

VI. Conclusion

We demonstrate that a lightweight DETR variant—with a reduced backbone and fewer transformer layers/queries—can outperform both CNN-only and Transformer-only models on UAV imagery while remaining feasible on mid-range GPUs. Although challenges with memory and transfer learning persist, this approach shows promise for efficient object detection in aerial contexts.

References

[1] WANDB: https://wandb.ai/DI_725_Assignment_2___2697258/DI_725_Assignment_2___2697258/reports/DI_725_Assignment_2____2697258--VmlldzoxMjM4NTg0MQ?accessToken=u5rbwmky6lg7pcpf16aavu89f3mewfsdc2twzgunqum380xw57tk9giq011ci5f8.

[2] GitHub: https://github.com/turgay-yildiz/DI-725-Transformers-and-Attention-Based-Deep-Networks-/tree/main/Assignment_2.